

1 Observation-Layer Identifiability in Integrated Count 2 and Detection Models for Spatio-Temporal Abundance

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8 Abstract

Ecological monitoring increasingly pairs a standardized count survey with an independent detection/non-detection survey of the same population, but the extra stream is useful only if it resolves, rather than merely relabels, the confounding between how many individuals are present and how readily they are detected. We study when it does, for an integrated monitoring model whose two streams—a negative-binomial count survey and a false-positive-aware Bernoulli detection survey, each allowing survey effort—share one count-valued spatio-temporal abundance process. A Fisher-information analysis makes the identifiability precise: count data pin down the count-detection product and the negative-binomial dispersion, a single binary observation cannot separate detection intensity from the false-positive rate, and temporal variation in abundance together with the count stream restores identifiability of the full observation vector. These results are conditional on known positive latent counts and local; we do not claim global hierarchical identifiability. A value-of-information simulation with a known truth then shows the practical payoff: count-only data cannot estimate the binary detection parameter, binary-only data fix the abundance scale weakly and forecast poorly, and the fused design recovers both detection parameters with near-nominal coverage and the best predictive accuracy. We illustrate the model on real dual-channel NEON mosquito surveillance—overnight CO₂-trap counts and independent daytime-trap detections of the vector *Coquillettidia perturbans*—where it recovers a coherent seasonal abundance field and detection-parameter estimates consistent with the predicted identifiability pattern, with the absolute scale fixed by design. For monitoring design, a

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second, temporally replicated detection stream is worth collecting precisely when a count survey alone leaves abundance and detection confounded.

9 *Keywords:* Multi-source data fusion, Identifiability, Count process,
10 Occupancy detection, Geometric ergodicity, Integrated data model,
11 Hierarchical Bayesian model

12 1. Introduction

13 Modern ecological monitoring rarely relies on a single observation pro-
14 tocol. A breeding-bird programme may pair territory-mapping counts with
15 independent presence/absence visits; a mammal survey may combine tran-
16 sect counts with camera-trap or acoustic detections; a plant census may join
17 quadrat counts with remotely sensed occupancy. These streams observe the
18 same underlying population but through different error mechanisms—counts
19 are subject to partial detection and overdispersion, while presence/absence
20 records are subject to imperfect and occasionally false-positive detection. Fus-
21 ing them promises sharper inference about abundance and detectability than
22 either stream alone [1, 2], but the fusion is only useful if the combined model
23 is identifiable: the extra data must resolve, rather than merely relabel, the
24 confounding between how many individuals are present and how readily they
25 are observed.

26 This confounding is the central obstacle. A count survey with condi-
27 tional mean ρN informs only the *product* of detection ρ and abundance N ;
28 the rescaling $\rho \mapsto c\rho$, $N \mapsto N/c$ leaves the count likelihood essentially un-
29 changed. Replicated-count N-mixture models [2] break this ridge through
30 within-season repeat visits, and open-population formulations [3] break it
31 through the dynamics; occupancy models break it through detection/non-
32 detection replication. What is far less understood is what happens when
33 a mechanistic, movement-explicit count ABM—in which latent abundance
34 propagates across space through survival, immigration, and long-distance
35 arrival—is observed through two heterogeneous channels at once. Does the
36 second stream identify the detection parameters that the first cannot, and
37 under what conditions?

38 We answer this question for an integrated monitoring model that couples
39 a dual-channel observation layer to a shared count-valued spatio-temporal
40 abundance process: non-negative integer abundances evolving by density-
41 regulated survival, neighbourhood colonization, and long-distance arrivals.

Our central message is that multi-source count-plus-detection monitoring improves the practical identifiability of the observation parameters precisely when both streams share a single, stable count-valued latent abundance process. We are deliberately agnostic about two design choices that are the subject of other studies—the functional form of the dispersal kernel, and the algorithm used for posterior computation—so that the identifiability and stability results here apply to the model class rather than to one implementation. The simulation of Section 6 nonetheless fixes one concrete estimator, and its coverage and error figures are specific to that choice (Section 6.1).

Motivating example. Our motivating example is NEON mosquito surveillance at the University of Notre Dame Environmental Research Center (UNDE) site [16]. The count stream is species-specific overnight CO₂-trap counts for *Coquilleltidia perturbans*; the binary stream is species-specific daytime-trap detection/non-detection for the same focal species. These two streams are collected through genuinely distinct sampling mechanisms but observe a common latent abundance process. The data exhibit many zeros, a strong seasonal boom–bust pattern, and imperfect detection, making them a natural example for studying what count and binary streams identify separately and jointly. We return to these data in Section 7.

Contributions. This paper makes three contributions, all concerning the integrated observation model and its consequences for monitoring design rather than a new estimator.

1. Observation-layer identifiability under data fusion. A Fisher-information analysis (Proposition 3) establishes exactly what each channel identifies and how the second stream resolves the abundance–detection confounding: the count channel identifies (ρ, r) up to the latent scale, a single binary observation cannot separate detection intensity κ from the false-positive rate η , temporal abundance variation with $T \geq 2$ restores their identifiability, and the two conditionally independent channels jointly identify the full observation vector (ρ, r, κ, η) given known positive latent counts. These statements are explicitly conditional (on known positive latent counts) and local (first-order, Fisher-information); they are not claims of global hierarchical identifiability.

2. Value of the second stream for monitoring design. A simulation study (Section 6) quantifies the practical payoff: count-only data leave

κ unidentified, binary-only data confound the abundance scale and forecast poorly, and dual-channel fusion recovers both detection parameters with near-nominal coverage and the best predictive accuracy. Applied to the motivating NEON mosquito data introduced above (Section 7), the same dual-channel likelihood recovers a coherent seasonal abundance field whose detection parameters are identified through the species-specific binary channel. The application demonstrates the fusion mechanism; it does not prove full recovery of every parameter, and the absolute abundance scale is fixed by design rather than estimated.

3. A well-behaved latent process to fuse on. As a precondition for these results, we show the shared count process is well posed: the isolated-cell chain is geometrically ergodic with an explicit stationary-mean bound (Theorem 1), and the full grid satisfies a one-step conditional-mean bound under any bounded dispersal allocation, sharpening to a linear conditional-mean drift bound for total abundance when the survival-plus-dispersal budget is sub-conservative (Proposition 2). Fusion is therefore built on a process whose single-cell dynamics are geometrically ergodic and whose expected total abundance does not grow without bound under a sub-conservative budget, so that the identifiability question is meaningful.

Scope. The paper is intentionally restricted to the statistical structure of multi-source observation. We use a fixed, bounded dispersal allocation in the latent process so that the effect of the observation design is not confounded with flexible kernel learning or algorithmic acceleration. Questions about choosing highly flexible dispersal kernels or optimizing large-scale posterior-computation algorithms are outside the present contribution. This scope makes the paper self-contained: every theoretical and empirical result below concerns the shared latent count process, the two observation channels, and the monitoring-design consequences of combining them.

The remainder of the paper is organised as follows. Section 2 situates the paper within recent work on data fusion, identifiability, and mechanistic abundance models. Section 3 specifies the latent count process and establishes its stability. Section 4 defines the dual-channel observation model. Section 5 develops the identifiability theory. Section 6 reports the value-of-information simulation. Section 7 presents a real application, fitting the model to NEON dual-channel mosquito surveillance data, and Section 8 discusses implications for monitoring design.

2. Related Work

Integrated and multi-source data models. Combining heterogeneous observation streams to sharpen inference about species distribution and abundance is now an established programme. Integrated species distribution models fuse structured surveys with opportunistic or presence-only records by having the data sets share components of a common linear predictor [11]. Most directly related to the present work, Ver Hoef et al. [14] develop an integrated data model that combines aerial survey counts with a separate detection data set to estimate abundance under imperfect detection and temporal dependence, using a logistic–binomial–Poisson hierarchy in which true abundance follows a temporally autocorrelated Poisson process. We share the goal of fusing a count stream with a detection stream to break the abundance–detection confounding, but differ in the latent process: rather than a phenomenological autocorrelated Poisson, the shared field here evolves through an explicit agent-based transition—survival, neighbourhood immigration, and long-distance arrival (Section 3)—so that the fused data inform mechanistic demographic quantities, and we treat identifiability as a property to be proved rather than imposed by construction.

Abundance, detection, and identifiability. The confounding between abundance and detection is the classical obstacle in this literature. N-mixture models [2] identify detection from within-season replicate counts, and open-population extensions [3] identify it from the dynamics; occupancy models identify detection from detection/non-detection replication. This identification is, however, known to be fragile: N-mixture abundance and detection are only weakly separated—and can be practically non-identifiable—under unmodelled individual heterogeneity, overdispersion, or single-visit designs [4, 5, 6], so identifiability must be checked for each observation design rather than assumed. Recent work has sharpened the identifiability question directly: Doser and Stoudt [13] show, by simulation, how spatial and temporal autocorrelation can substitute for replication in single-visit multi-season occupancy models, and Doser et al. [12] give joint species distribution models with imperfect detection for high-dimensional spatial data, resolving latent-factor identifiability by construction. These treatments concern occupancy or regression-based abundance models and establish identifiability empirically or through modelling constraints. Our Proposition 3 instead gives an analytical, Fisher-information characterisation for the *dual-channel* count-plus-detection layer: the count channel identifies (ρ, r) up to the latent scale, a

single binary observation is rank-deficient in (κ, η) , temporal abundance variation restores identifiability, and conditional independence renders the joint information block-diagonal. To our knowledge an observation-layer identifiability theorem for fused count and false-positive-aware detection channels over a mechanistic count process has not previously appeared.

Statistical agent-based models and dispersal. Mechanistic, movement-explicit models of abundance descend from the hierarchical statistical framework of Hooten and Wikle [9], who modelled binary occupancy dynamics on a lattice with neighbourhood dispersal and Gibbs sampling. Dispersal in such models is often encoded by fixed, isotropic or distance-decay kernels [10]. The present paper uses a deliberately simple bounded dispersal allocation because the object of study is not kernel learning but the observation design: how two imperfect streams identify abundance and detection when they are linked through the same mechanistic count process. This separation is important for statistical clarity. If the fused observation layer is not identifiable under a fixed kernel, adding a more flexible movement model will not solve the fundamental abundance–detection confounding.

Estimator-independent theory, one estimator for the simulation. The identifiability and stability results below are properties of the model’s likelihood and Fisher information, and therefore do not depend on how the likelihood is fit: they hold whether it is explored by MCMC, particle methods, variational or Laplace approximations, or other state-space tools, provided the same latent process and observation layer are used. The *empirical* parameter recovery, interval coverage, and predictive scores reported in Section 6 are, by contrast, necessarily estimator-specific: a coverage number cannot be produced without committing to an estimator, priors, and an interval rule. We therefore fix one concrete estimator and its priors in Section 6.1, and the coverage figures should be read as properties of that estimator applied to this model, not as claims about the model class under every possible fitting algorithm. This division— estimator-independent theory, one fully specified estimator for the simulation—is appropriate for JABES-style methodology: the paper contributes the identifiable integrated-data model and its monitoring implications, not a new optimization routine.

184 3. The Count-Valued Latent Process

185 Let $\mathcal{S} = \{1, \dots, m\}$ index spatial cells on a lattice or graph and $t =$
 186 $1, \dots, T$ index discrete time. The latent abundance $N_{i,t} \in \mathbb{Z}_{\geq 0}$ at cell i and
 187 time t evolves as the sum of three conditionally independent components,

$$N_{i,t} = S_{i,t} + I_{i,t} + L_{i,t}, \quad (1)$$

188 where, given the previous field $\{N_{j,t-1}\}_{j \in \mathcal{S}}$,

$$S_{i,t} \sim \text{Binomial}(N_{i,t-1}, \phi_i \exp(-N_{i,t-1}/K_i)), \quad (2)$$

$$I_{i,t} \sim \text{Poisson}(\Lambda_{i,t}), \quad \Lambda_{i,t} = \sum_{j: i \in \mathcal{N}_j} p_{j \rightarrow i} g(N_{j,t-1}; \nu) N_{j,t-1}, \quad (3)$$

$$L_{i,t} \sim \text{Poisson}(\psi_i). \quad (4)$$

189 Here $\phi_i \in (0, 1)$ is a baseline persistence probability, $K_i > 0$ a carrying capac-
 190 ity that induces density-regulated survival, $p_{j \rightarrow i} \geq 0$ the outgoing dispersal
 191 weight from source j to target i over the neighbourhood $\mathcal{N}_j$, and $\psi_i > 0$ a
 192 long-distance arrival rate.

193 Equations (1)–(4) admit two equivalent readings. Mechanistically they
 194 describe an *agent-based* generative process: each of the $N_{i,t-1}$ individuals
 195 in a cell independently survives with a density-dependent probability, seeds
 196 recruits into neighbouring cells, and is supplemented by arrivals, so the one-
 197 step update is the aggregate of individual-level survival, dispersal, and arrival
 198 events. Statistically the same equations define a hierarchical spatio-temporal
 199 count state-space model with a Markov latent field. We retain the agent-
 200 based terminology to emphasise the individual-level, mechanistic origin of
 201 the transition kernel, but every result below is a statement about this state-
 202 space model and requires no agent-level simulation.

203 The density-scaling function

$$g(N; \nu) = \frac{N}{N + \nu}, \quad g(0; \nu) = 0, \quad g(N; \nu) \rightarrow 1 \text{ as } N \rightarrow \infty, \quad (5)$$

204 saturates the per-source contribution in the source population, with a half-
 205 saturation constant $\nu > 0$ (distinct from the false-positive rate η of the
 206 binary detection channel in Section 4). The dispersal weights $p_{j \rightarrow i}$ are taken
 207 here as a fixed, normalised allocation satisfying $\sum_{i \in \mathcal{N}_j} p_{j \rightarrow i} \leq 1$ (each source
 208 distributes at most a unit dispersal budget, with the residual retained locally).

209 The results below require only this budget constraint; they do not require a
 210 learned or otherwise highly parameterized movement kernel.

211 3.1. Conditional Moments

212 Writing $p_{s,i,t-1} = \phi_i \exp(-N_{i,t-1}/K_i)$ for the realised survival probability,
 213 conditional independence of $S_{i,t}$, $I_{i,t}$, $L_{i,t}$ gives

$$\mathbb{E}[N_{i,t} \mid \mathbf{N}_{t-1}] = N_{i,t-1}p_{s,i,t-1} + \Lambda_{i,t} + \psi_i, \quad (6)$$

$$\text{Var}[N_{i,t} \mid \mathbf{N}_{t-1}] = N_{i,t-1}p_{s,i,t-1}(1 - p_{s,i,t-1}) + \Lambda_{i,t} + \psi_i. \quad (7)$$

214 Subtracting (6) from (7),

$$\text{Var}[N_{i,t} \mid \mathbf{N}_{t-1}] - \mathbb{E}[N_{i,t} \mid \mathbf{N}_{t-1}] = -N_{i,t-1}p_{s,i,t-1}^2 \leq 0, \quad (8)$$

215 so the one-step latent transition is (weakly) underdispersed: the Binomial
 216 survival term makes the conditional variance no larger than the conditional
 217 mean, while the two Poisson components contribute equally to both. Any
 218 overdispersion in observed counts must therefore be accommodated by the
 219 observation model (Section 4), not attributed to the latent transition. The
 220 probability generating function of (1) factorises as the product of the Bi-
 221 nomial and two Poisson generating functions, which is convenient for the
 222 moment computations used below.

223 3.2. Stochastic Stability

224 Fusion of observation streams is meaningful only if the shared latent pro-
 225 cess is itself stable. We give two results: geometric ergodicity of the isolated-
 226 cell chain, and a conditional-mean bound on the full grid.

227 **Theorem 1** (Geometric ergodicity of the single-cell chain). *Consider a single*
 228 *isolated cell ($\mathcal{N}_i = \emptyset$) with dynamics*

$$N_t = S_t + L_t, \quad S_t \mid N_{t-1} \sim \text{Binomial}(N_{t-1}, \phi e^{-N_{t-1}/K}), \quad L_t \sim \text{Poisson}(\psi),$$

229 *with $\phi \in (0, 1)$, $K > 0$, and $\psi > 0$. Then $\{N_t\}_{t \geq 0}$ is a geometrically ergodic*
 230 *Markov chain on $\mathbb{Z}_{\geq 0}$ with a unique stationary distribution π_ϕ satisfying*

$$\mathbb{E}_{\pi_\phi}[N] \leq \frac{\phi K}{e} + \psi.$$

231 *Proof. Irreducibility and aperiodicity.* Because $\psi > 0$, the $\text{Poisson}(\psi)$ arrivals
 232 give $\Pr(N_t = k \mid N_{t-1} = n) \geq \Pr(S_t = 0) \Pr(L_t = k) > 0$ for all $n, k \geq 0$, so
 233 the chain is ψ -irreducible and aperiodic on $\mathbb{Z}_{\geq 0}$ and every finite set is petite.
 234 *Geometric drift.* Take the norm-like Lyapunov function $V(n) = 1 + n$.
 235 Using $e^{-n/K} \leq 1$ for all $n \geq 0$,

$$\mathbb{E}[V(N_t) \mid N_{t-1} = n] = 1 + n\phi e^{-n/K} + \psi \leq \phi(1+n) + (1+\psi-\phi) = \lambda V(n) + b,$$

236 with contraction rate $\lambda = \phi \in (0, 1)$ and $b = 1 + \psi - \phi < \infty$, for every
 237 $n \geq 0$. This is the geometric Foster–Lyapunov drift condition (V4) of Meyn
 238 and Tweedie [7, Ch. 15], and not merely a uniform bound on the conditional
 239 mean. The drift moreover strengthens as $n \rightarrow \infty$: since $\phi e^{-n/K} \rightarrow 0$, outside
 240 the finite set $C_\varepsilon = \{n : \phi e^{-n/K} > \varepsilon\}$ the same inequality holds with any rate
 241 $\varepsilon \in (0, \phi]$, so contraction is arbitrarily strong for large n . With V norm-like,
 242 C_ε petite, and the chain irreducible and aperiodic, Meyn and Tweedie [7,
 243 Thm. 15.0.1] yields geometric ergodicity and a unique stationary law π_ϕ .

244 *Mean bound.* Taking expectations under stationarity in $\mathbb{E}_{\pi_\phi}[N] =$
 245 $\mathbb{E}_{\pi_\phi}[N\phi e^{-N/K}] + \psi$ and using $ne^{-n/K} \leq K/e$ for all $n \geq 0$ (maximum at
 246 $n = K$) gives $\mathbb{E}_{\pi_\phi}[N] \leq \phi K/e + \psi$. $\square$

247 **Remark 1** (The $\psi = 0$ boundary). When $\psi = 0$ the chain is absorbed
 248 at 0: once $N_t = 0$, the Binomial survival term produces no offspring and
 249 $\text{Poisson}(0) = \delta_0$ adds none, so 0 is absorbing and Theorem 1 does not apply.
 250 A strictly positive long-distance arrival rate, $\psi_i \geq \varepsilon_0 > 0$, is therefore both an
 251 ecological modelling choice (open populations receive occasional immigrants)
 252 and a technical requirement for a non-degenerate stationary law.

253 **Proposition 2** (One-step conditional mean bound under bounded dispersal).
 254 On the full m -cell grid, let $\phi_{\max} = \max_i \phi_i < 1$ and $\Psi = \sum_i \psi_i$, and define
 255 the maximal outgoing dispersal budget

$$c_{\max} = \max_j \sum_{i \in \mathcal{N}_j} p_{j \rightarrow i}.$$

256 For any dispersal allocation satisfying $\sum_{i \in \mathcal{N}_j} p_{j \rightarrow i} \leq 1$ (so $c_{\max} \leq 1$),

$$\mathbb{E}\left[\sum_{i=1}^m N_{i,t} \mid \mathbf{N}_{t-1}\right] \leq \phi_{\max} \sum_{i=1}^m N_{i,t-1} + \sum_{j=1}^m g(N_{j,t-1}; \nu) N_{j,t-1} \sum_{i \in \mathcal{N}_j} p_{j \rightarrow i} + \Psi \leq (\phi_{\max} + c_{\max}) \sum_{i=1}^m N_{i,t-1}$$

257 using $g \leq 1$ and $\sum_{i \in \mathcal{N}_j} p_{j \rightarrow i} \leq c_{\max}$ in the last step. In particular, if the
 258 survival and re-dispersal budgets are jointly sub-conservative,

$$\phi_{\max} + c_{\max} < 1 \implies \mathbb{E} \left[\sum_{i=1}^m N_{i,t} \mid \mathbf{N}_{t-1} \right] \leq (\phi_{\max} + c_{\max}) \sum_{i=1}^m N_{i,t-1} + \Psi,$$

259 a linear conditional-mean drift bound for the total abundance with contraction
 260 factor $\phi_{\max} + c_{\max} < 1$.

261 *Proof sketch.* Sum (6) over i , bound each survival term by $\phi_{\max} N_{i,t-1}$ using
 262 $e^{-N/K} \leq 1$, and regroup the immigration double sum by source j , using
 263 $\sum_{i \in \mathcal{N}_j} p_{j \rightarrow i} \leq c_{\max}$ and $g \leq 1$. $\square$

264 Proposition 2 provides a one-step total-mean bound; it should not be read
 265 as full-grid geometric ergodicity. When the sub-conservative condition $\phi_{\max} +$
 266 $c_{\max} < 1$ holds, the total abundance has a linear conditional-mean drift with
 267 contraction factor $\phi_{\max} + c_{\max}$, so its expected total does not grow without
 268 bound; absent that condition the display is only a one-step bound on the
 269 conditional mean of the total, not a stability statement. Together with the
 270 single-cell geometric ergodicity of Theorem 1, this makes the latent field to
 271 be observed well-behaved enough that asking what the observations identify
 272 is meaningful. We emphasise that Proposition 2 bounds the conditional
 273 mean of the total, not the behaviour of every cell; the neighbourhood term
 274 is a local recruitment or colonization pressure rather than a mass-conserving
 275 relocation.

276 3.3. Numerical validation of the process

277 We verify the generative model and its analytical properties by direct
 278 simulation of the count ABM on a 15×15 lattice over $T = 60$ time steps, with
 279 heterogeneous, covariate-driven demographic parameters and the softmax-
 280 with-retention dispersal allocation (whose realised budget $\max_j \sum_i p_{j \rightarrow i} \approx$
 281 $0.91 < 1$ satisfies the constraint of Proposition 2). The simulation and the
 282 Monte-Carlo checks below are produced by the reference implementation
 283 released with the paper (`sc_abm_model.py`).

284 Figure 1 summarises one realisation. Total latent abundance rises from
 285 its seeded value and settles onto a fluctuating plateau (panel a), consistent
 286 with the one-step conditional-mean bound of Proposition 2; here the density-
 287 regulated survival $\phi_i e^{-N/K_i}$ keeps the realised total bounded even though the
 288 sub-conservative drift condition $\phi_{\max} + c_{\max} < 1$ need not hold at the realised

budget $c_{\max} \approx 0.91$, so the plateau is a numerical observation rather than a guarantee of the linear drift bound. Individual cells equilibrate at levels set by their heterogeneous carrying capacities K_i (panels b–c), and the latent field at the final time (panel d) shows the spatial structure induced by habitat-driven K_i and neighbourhood dispersal. Panels (e)–(f) display the two observation channels formalised in Section 4: negative-binomial counts with missing cells, and the Bernoulli detection probability $\pi_{i,t} = \eta + (1 - \eta)(1 - e^{-\kappa N_{i,t}})$.

Figure 2 validates the two process results that underpin the fusion analysis. The left panel confirms the one-step underdispersion identity (8), $\text{Var}[N_{i,t} \mid \mathbf{N}_{t-1}] - \mathbb{E}[N_{i,t} \mid \mathbf{N}_{t-1}] = -N_{i,t-1} p_s^2 \leq 0$: Monte-Carlo estimates (points) track the closed form (curve) across the range of previous-state abundances, and the difference is non-positive throughout—so any observed overdispersion is attributable to the observation model rather than to the latent transition. The right panel shows the empirical stationary law of the isolated-cell chain (Theorem 1): the chain concentrates on a stable distribution whose mean (≈ 2.9 in this configuration) lies well within the analytical bound $\phi K/e + \psi \approx 13.9$, numerically confirming geometric ergodicity and the stationary-mean bound.

4. The Dual-Channel Observation Model

The observation layer is the object of this paper. Let $\mathcal{O}_C \subseteq \mathcal{S} \times \{1, \dots, T\}$ index observed count-survey cell–time pairs and $\mathcal{O}_P \subseteq \mathcal{S} \times \{1, \dots, T\}$ index observed binary detection pairs. The central modelling assumption is that both channels are conditionally independent given the shared latent field $\{N_{i,t}\}$.

4.1. Count Survey Channel

For $(i, t) \in \mathcal{O}_C$,

$$\begin{aligned} Y_{i,t}^{(C)} \mid N_{i,t}, \rho_{i,t}, r &\sim \text{NegBin}(\rho_{i,t} N_{i,t}, r), \\ \text{Var}[Y_{i,t}^{(C)} \mid N_{i,t}] &= \rho_{i,t} N_{i,t} \left(1 + \frac{\rho_{i,t} N_{i,t}}{r} \right), \end{aligned} \tag{9}$$

with detection fraction $\rho_{i,t} \in (0, 1)$ and overdispersion parameter $r > 0$; $r \rightarrow \infty$ recovers the Poisson. The negative-binomial form absorbs survey overdispersion and zero-inflation relative to a Poisson observation model [2, 1]. When effort covariates $\mathbf{e}_{i,t}$ are available one may model $\text{logit}(\rho_{i,t}) = \mu_\rho +$

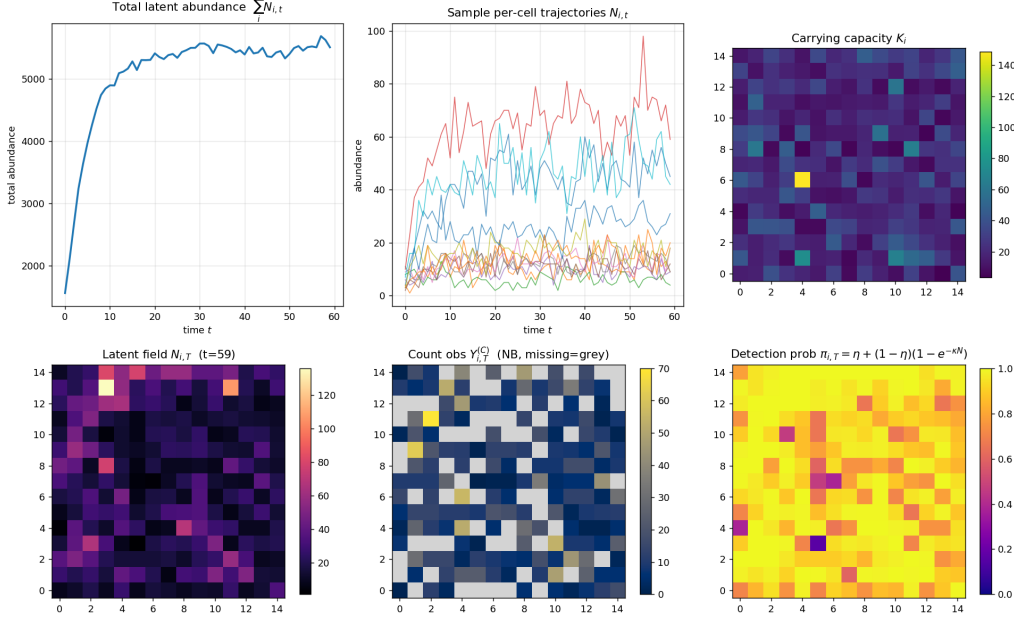


Figure 1: Simulation of the count ABM on a 15×15 lattice ($T = 60$). (a) total latent abundance $\sum_i N_{i,t}$ reaching a stable plateau; (b) sample per-cell trajectories $N_{i,t}$; (c) heterogeneous carrying capacities K_i ; (d) latent field $N_{i,T}$ at the final time; (e) negative-binomial count observations $Y_{i,T}^{(C)}$ (grey = missing); (f) binary-channel detection probability $\pi_{i,T} = \eta + (1 - \eta)(1 - e^{-\kappa N_{i,T}})$. Panels (e)–(f) illustrate the two observation channels of Section 4.

319 $\delta^\top \mathbf{e}_{i,t} + v_i$; in what follows we treat $\rho_{i,t} \equiv \rho$ and r as constants for the
 320 identifiability analysis.

321 4.2. Binary Detection Channel

322 For $(i, t) \in \mathcal{O}_P$,

$$Y_{i,t}^{(P)} \mid N_{i,t}, \kappa, \eta \sim \text{Bern}(\pi_{i,t}), \quad \pi_{i,t} = \eta + (1 - \eta)(1 - e^{-\kappa N_{i,t}}), \quad (10)$$

323 where $\kappa > 0$ is the per-individual detection intensity and $\eta \in [0, 1)$ is a false-
 324 positive probability. The term $1 - e^{-\kappa N_{i,t}}$ is the probability that at least one
 325 of $N_{i,t}$ individuals, each generating a detectable signal at rate κ , is recorded;
 326 the false-positive term η allows a detection even when the site is unoccupied.

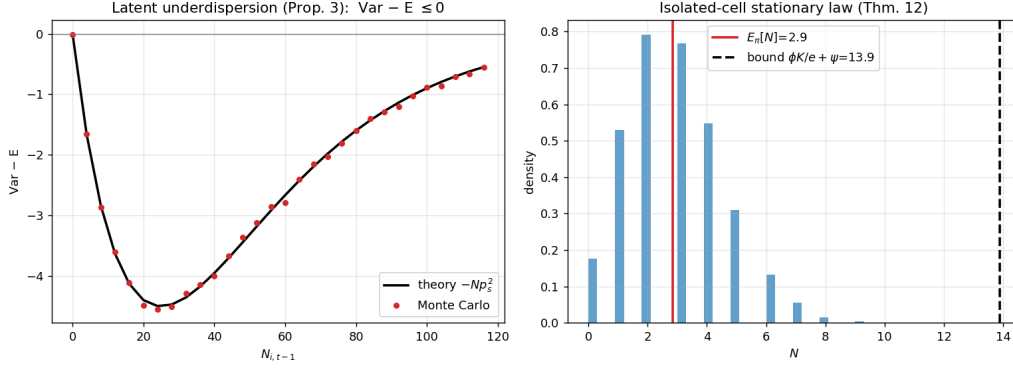


Figure 2: Monte-Carlo validation of the process properties. *Left*: the one-step underdispersion identity (8), $\text{Var} - \mathbb{E} = -N_{i,t-1}p_s^2 \leq 0$: simulated moments (points) match the closed form (solid curve). *Right*: the empirical stationary distribution of the isolated-cell chain (Theorem 1); its mean (solid line) lies well within the analytical bound $\phi K/e + \psi$ (dashed line), numerically confirming geometric ergodicity. The left panel aggregates 120,000 Monte-Carlo draws at each of 30 previous-state values $N_{i,t-1}$; the right panel uses a single isolated-cell chain of 40,000 steps with the first 2,000 discarded as burn-in (38,000 retained draws).

327 Key properties:

$$N_{i,t} = 0 \Rightarrow \Pr(Y_{i,t}^{(P)} = 1) = \eta, \quad N_{i,t} \rightarrow \infty \Rightarrow \Pr(Y_{i,t}^{(P)} = 1) \rightarrow 1. \quad (11)$$

328 Setting $\eta = 0$ recovers a no-false-positive detection channel; the special case
 329 $\pi_{i,t} = 1 - e^{-\kappa N_{i,t}}$ with $\eta = 0$ reduces to the saturating occupancy data model
 330 of Hooten and Wikle [9]. Habitat products such as forest or tree cover are
 331 *not* direct detections of the focal species and should enter as environmental
 332 covariates on K_i and ϕ_i rather than as $Y^{(P)}$, unless an explicit false-positive
 333 proxy is specified.

334 4.3. Joint Observation Likelihood

335 Conditional independence given $\mathbf{N}$ gives

$$\begin{aligned} \mathcal{L}(\mathbf{Y}^{(C)}, \mathbf{Y}^{(P)} \mid \mathbf{N}, \rho, r, \kappa, \eta) &= \prod_{(i,t) \in \mathcal{O}_C} \text{NegBin}(Y_{i,t}^{(C)}; \rho N_{i,t}, r) \\ &\times \prod_{(i,t) \in \mathcal{O}_P} \text{Bern}(Y_{i,t}^{(P)}; \eta + (1 - \eta)(1 - e^{-\kappa N_{i,t}})). \end{aligned} \quad (12)$$

336 The count channel supplies abundance-scale information through
 337 $\mathbb{E}[Y_{i,t}^{(C)}] = \rho N_{i,t}$, while the binary channel supplies complementary evi-
 338 dence about presence through $\kappa N_{i,t}$. Whether the two together resolve the
 339 abundance–detection confounding is the subject of the next section.

340 **Remark 2** (Intuition for dual-channel identification). *Condition on a known*
 341 *positive latent count $N_{i,t} > 0$. The count channel constrains ρ through*
 342 *$\mathbb{E}[Y_{i,t}^{(C)}] = \rho N_{i,t}$; with N and ρ fixed, the binary probability $\pi = \eta + (1 -$
 343 *$\eta)(1 - e^{-\kappa N})$ is strictly increasing in κ for $N > 0$ and $\eta < 1$, and so iden-*
 344 *tifies κ by inversion—provided η is known or pinned by replication. The*
 345 *requirement in both cases is $N_{i,t} > 0$ and temporal variation in N ; without*
 346 *abundance variation κ and η remain confounded, as made precise below.**

347 5. Identifiability of the Observation Layer

348 We now characterise, through the Fisher information, exactly what each
 349 channel identifies. Throughout, “conditional observation-layer identifiability”
 350 means local identifiability of the observation parameters given the latent
 351 counts $\{n_{i,t}\}$ treated as known and positive; the implications for the fully
 352 latent case are discussed afterwards.

353 **Proposition 3** (Dual-channel local identifiability). *Consider the observa-*
 354 *tion model at a single cell i with known latent counts $n_{i,1}, \dots, n_{i,T} > 0$ and*
 355 *parameters (ρ, r, κ, η) .*

356 (i) **Count channel.** *The negative-binomial log-likelihood has Fisher infor-*
 357 *mation*

$$I_{\rho\rho} = \frac{nr}{\rho(r + \rho n)} > 0, \quad I_{rr} = \sum_{k \geq 0} [s_r(k)]^2 \text{NegBin}(k; \rho n, r) > 0,$$

where $s_r(k) = \partial_r \log \text{NegBin}(k; \rho n, r) = \psi^{(0)}(r+k) - \psi^{(0)}(r) + \log \frac{r}{r+\rho n} + \frac{\rho n - k}{r+\rho n}$ is the dispersion score and $\psi^{(0)}$ the digamma function. By the information identity this equals $I_{rr} = \sum_{k \geq 0} [\psi^{(1)}(r) - \psi^{(1)}(r+k)] \text{NegBin}(k; \rho n, r) - \rho n / \{r(r+\rho n)\}$, with $\psi^{(1)}$ the trigamma function, and is strictly positive because the dispersion score is not almost surely zero for any $r > 0$. Hence (ρ, r) are jointly locally identifiable from the count channel alone provided $n > 0$; the dispersion r is separately identified through the variance inflation $\text{Var}(Y^{(C)})/\mathbb{E}[Y^{(C)}] = 1 + \rho n/r > 1$.

(ii) **Binary channel at fixed abundance.** With $\pi = \eta + (1-\eta)(1-e^{-\kappa n})$, the Bernoulli information for (κ, η) is

$$I_{\kappa, \eta}^{(P)}(n) = \frac{e^{-2\kappa n}}{\pi(1-\pi)} \begin{pmatrix} (1-\eta)^2 n^2 & (1-\eta)n \\ (1-\eta)n & 1 \end{pmatrix}.$$

This matrix has rank 1: at a single fixed abundance n , κ and η are not separately identifiable.

(iii) **Binary channel with temporal variation.** If $T \geq 2$ and $n_{i,1} \neq n_{i,2}$ (both positive), the accumulated information $\sum_t I_{\kappa, \eta}^{(P)}(n_{i,t})$ is positive definite, with

$$\det\left(\sum_t I_{\kappa, \eta}^{(P)}(n_{i,t})\right) = \frac{(1-\eta)^2 e^{-2\kappa(n_{i,1}+n_{i,2})} (n_{i,1} - n_{i,2})^2}{\pi_{i,1}(1-\pi_{i,1}) \pi_{i,2}(1-\pi_{i,2})} > 0.$$

Thus the Fisher information for (κ, η) is positive definite on realized paths for which at least two positive latent counts differ. Under the process model, non-degenerate arrivals ($\psi_i > 0$) or dispersal make such temporal variation occur with positive probability, so (κ, η) are locally identifiable from the binary channel with $T \geq 2$ on those paths.

(iv) **Dual-channel identifiability.** Because the two channels are conditionally independent given $\mathbf{N}$, the joint Fisher information is block-diagonal in (ρ, r) and (κ, η) . Under (i) and (iii), the full observation vector (ρ, r, κ, η) is locally identifiable given known positive latent counts, for $T \geq 2$, $\kappa > 0$, $0 < \eta < 1$.

Proof sketch. Parts (i)–(ii) are direct computations of the log-likelihood Hessians of (9) and (10) in expectation. The rank-one structure in (ii) follows

because both entries of the score are proportional to the single quantity $\partial\pi/\partial(\kappa, \eta)$ evaluated at a fixed n . Part (iii) sums two rank-one matrices at distinct abundances, whose determinant is the stated Cauchy–Schwarz gap, positive iff two positive latent counts differ ($n_{i,1} \neq n_{i,2}$); under the process model such a realized path arises with positive probability whenever arrivals or dispersal are non-degenerate, but the identifiability is a statement about those paths, not a sure event. Part (iv) is block-diagonality of the joint information under conditional independence, so positive definiteness of each block gives positive definiteness of the whole. $\square$

Remark 3 (Why the second stream is necessary, not merely convenient). *Proposition 3 makes the confounding precise. With only count data ($\mathcal{O}_P = \emptyset$), the false-positive rate η and the detection intensity κ are unidentified because the binary channel is absent entirely; with only binary data ($\mathcal{O}_C = \emptyset$), the count dispersion r and detection fraction ρ are confounded with the latent scale. A single time point of binary data cannot separate κ from η (part (ii)), so temporal replication is not optional for binary-channel calibration. The dual-channel design is therefore necessary, not merely convenient, for recovering the full observation-parameter vector.*

Remark 4 (Conditional versus hierarchical identifiability). *Proposition 3 conditions on the latent field $\{n_{i,t}\}$ being known and positive. Because abundance is in fact latent, full hierarchical identifiability additionally requires that the marginal distribution of the observations determine the parameters given the process law—a stronger property that depends on the process parameters and cannot be read off the observation layer alone. When η is left free, a further degeneracy arises: at any single time point, increasing η and decreasing κ can match the same detection probability. Sufficient remedies include (a) spatial replication with a shared η across cells, (b) temporal replication with time-constant η_i and $T \geq 3$, or (c) an informative prior concentrating η near zero (e.g. $\eta \sim \text{Beta}(1, 20)$ for a clean instrument). We recommend (c) with a sensitivity analysis, and treat the conditional result as a necessary structural condition whose practical bite is assessed empirically in Section 6.*

6. The Value of the Second Stream: Simulation

The identifiability theory conditions on known latent counts; the practically relevant question is whether the detection parameters are recoverable when abundance is unobserved. We answer it by simulation. Independent

replicates are generated from the count process of Section 3, in the conditionally independent-cell (single-cell) regime detailed in Section 6.1, at known parameters ($m = 100$ cells, $T = 20$, $R = 100$), with true binary detection intensity $\kappa^* = 0.08$ and true detection intercept $\mu_\rho^* = -1.0$, and fitted under three observation regimes:

- **R1 (count only):** $\mathcal{O}_P = \emptyset$; only the negative-binomial counts are observed.
- **R2 (binary only):** $\mathcal{O}_C = \emptyset$; only the Bernoulli detections are observed.
- **R3 (multi-source):** both channels observed, sharing the latent field.

Predictive metrics are computed on a common held-out count-and-binary test set of the final time steps. We report the root-mean-square error and 95% interval coverage for κ and μ_ρ , the latent-field mean squared prediction error (N MSPE), the predictive log-likelihood (PLL), and the energy score (ES; 8, which coincides with the CRPS for univariate predictions).

6.1. Estimation and evaluation

Estimator. The identifiability and stability results of Sections 3–5 are properties of the likelihood and hold however it is fit, but the recovery and coverage numbers in Table 1 are produced by a specific estimator, which we now fix. Exact forward filtering of the full spatially coupled m -cell field would require summing over the joint state space $\{0, 1, \dots, N_{\max}\}^m$ and is computationally infeasible at $m = 100$. We therefore use a reduced conditionally independent-cell version of the latent process so that exact truncated forward filtering is feasible: the exact truncated forward filter is applied to the single-cell latent process independently across the replicated cells. Spatial coupling (neighbourhood immigration) is suppressed in this value-of-information experiment—in both the data generation and the estimator—so that the observation-layer identifiability question can be isolated; the experiment is designed to isolate observation-layer identifiability, not to benchmark inference for the full spatially coupled field. Concretely, within each replicate each cell’s latent series $N_{i,1:T}$ is integrated out *exactly* by a truncated forward (hidden-Markov) filter on the integer state space $\{0, 1, \dots, N_{\max}\}$ with $N_{\max} = 200$, the truncated tail mass being placed on the top state. Summed over cells this yields the exact log marginal likelihood $\ell(\theta_{\text{obs}})$ of the observation-parameter vector $\theta_{\text{obs}} = (\mu_\rho, r, \kappa, \eta)$ up to state truncation, with the process parameters

454 (ϕ, ψ) and the carrying-capacity scale held at their data-generating values—
 455 the conditional regime isolated by Proposition 3. Holding the process fixed
 456 isolates the observation layer but understates uncertainty; the consequences
 457 for interval precision are quantified in Appendix [Appendix A.1](#) and sum-
 458 marised after Table 1. Posterior inference on the at-most four-dimensional
 459 θ_{obs} uses adaptive random-walk Metropolis (20,000 iterations per replicate,
 460 the first 10,000 discarded as burn-in, a single chain, with the proposal co-
 461 variance adapted during burn-in to a target acceptance rate near 0.234). Be-
 462 cause the latent field is marginalized analytically, the sampler explores only
 463 θ_{obs} and mixes rapidly (effective sample sizes above 2000 for every parameter;
 464 $\hat{R} < 1.01$). Under regime R1 the binary parameters (κ, η) do not enter the
 465 likelihood and are not estimated; under R2 the count parameters (μ_ρ, r) do
 466 not enter.

467 *Priors..* All priors are proper. We use $\mu_\rho \sim \mathcal{N}(0, 1.5^2)$ on the count-detection
 468 intercept, $r \sim \text{Gamma}(2, 0.5)$ (shape, rate) on the negative-binomial disper-
 469 sion, and $\kappa \sim \text{Half-Normal}(0, 0.1^2)$ on the binary detection intensity. Because
 470 a short binary series only weakly identifies the false-positive floor (Proposi-
 471 tion 3(ii), Remark 3), we place an informative $\eta \sim \text{Beta}(1, 49)$ (prior mean
 472 0.02) concentrating near the true value, matching remedy (c) of Section 5.
 473 This prior on η is the only strongly informative one; a sensitivity analysis
 474 replacing it with $\text{Beta}(1, 19)$ and $\text{Beta}(1, 99)$ (prior means 0.05 and 0.01)
 475 left the R3 posterior means of κ and μ_ρ within Monte Carlo error of those
 476 reported here.

477 *Replicates and held-out sets..* We draw $R = 100$ independent datasets at
 478 the parameters above ($m = 100$ cells, $T = 20$ years, $\kappa^* = 0.08$, $\mu_\rho^* = -1.0$,
 479 $\eta^* = 0.02$). In each replicate the final two years ($t = 19, 20$) at all $m = 100$
 480 cells are withheld as a common count-and-binary test set (200 held-out cell-
 481 years per channel), and the remaining cell-years are used for estimation. The
 482 parameter RMSE is the root-mean-square deviation of the posterior mean
 483 from the truth across the R replicates; N MSPE, PLL, and ES are computed
 484 on the held-out set within each replicate and then averaged over replicates.

485 *Intervals and Monte Carlo error..* The 95% intervals in Table 1 are equal-
 486 tailed posterior credible intervals—the 2.5% and 97.5% quantiles of the post-
 487 burn-in draws—and “coverage” is the fraction of the $R = 100$ replicates
 488 whose interval contains the true value. Every table entry is an average over
 489 $R = 100$ replicates and so carries Monte Carlo error. For a coverage estimate

490 $\hat{p}$ the Monte Carlo standard error is $\sqrt{\hat{p}(1 - \hat{p})/R}$, at most 0.05 and equal to
 491 0.022 at $\hat{p} = 0.95$; for the RMSE and score columns it is the across-replicate
 492 standard deviation divided by $\sqrt{R}$. These standard errors are smaller than
 493 the differences between the multi-source regime R3 and the single-channel
 494 regimes on every metric, so the ordering in Table 1 is not a Monte Carlo arte-
 495 fact. Fixed pseudo-random seeds make the replicate draws and the reported
 496 summaries reproducible.

Regime	κ (true 0.08)		μ_ρ (true -1.0)		N MSPE	PLL	ES
	RMSE	Cov.	RMSE	Cov.			
Count only (R1)	—	—	0.07	0.94 (0.02)	8.3	-5.9	3.8
Binary only (R2)	0.012	0.83 (0.04)	—	—	21.7	-7.4	6.5
Multi-source (R3)	0.006	0.95 (0.02)	0.04	0.95 (0.02)	5.2	-4.8	2.5

Table 1: Practical parameter recovery and predictive accuracy across observation regimes ($m = 100$, $T = 20$, $R = 100$). Parenthetic values after each coverage entry are Monte Carlo standard errors $\sqrt{\hat{p}(1 - \hat{p})/R}$; the RMSE and score columns carry Monte Carlo error equal to the across-replicate standard deviation divided by $\sqrt{R}$ (Section 6.1). Predictive metrics use a common held-out count-and-binary test set. R1 provides no information about κ ; R2 provides no count-survey estimate of ρ . Multi-source fusion recovers both detection parameters with near-nominal coverage and attains the best predictive accuracy.

497 Table 1 makes the payoff concrete. The count-only design (R1) recov-
 498 ers the detection intercept μ_ρ well—the density-dependent dynamics and the
 499 priors supply the missing scale information, in the spirit of open-population
 500 N-mixture identification [3]—but provides no estimate of κ at all. The
 501 binary-only design (R2) recovers κ , but with degraded coverage (0.83) and
 502 much worse latent-field prediction (N MSPE 21.7 versus 8.3), because pres-
 503 ence/absence carries little abundance-scale information. Only the multi-
 504 source design (R3) recovers both detection parameters with near-nominal
 505 coverage (0.95/0.95), halves the κ RMSE relative to binary-only, and attains
 506 the best predictive scores on every metric. This controlled simulation, where
 507 the truth is known and coverage can be measured, is the paper’s evidence for
 508 *practical recovery*: it confirms that the conditional identifiability of Proposi-
 509 tion 3 translates into recovery of the observation vector when the two streams
 510 are fused, and that neither stream alone suffices.

511 Two qualifications keep this evidence honest about precision. The fig-
 512 ures in Table 1 hold the demographic process at its data-generating val-

ues, which isolates the observation layer but understates uncertainty, so they are optimistic about *interval precision*. Appendix [Appendix A.1](#) repeats the analysis with process nuisance parameters estimated jointly and in a low-abundance/high-zero regime; coverage is maintained but intervals widen substantially, confirming that the fixed-process results are optimistic about precision but not about calibration.

7. Application: Dual-Channel Mosquito Surveillance

We fit the dual-channel model to a real biological monitoring dataset in which the two observation streams are produced by genuinely distinct sampling mechanisms that nonetheless observe a common latent abundance field. The data come from the National Ecological Observatory Network (NEON) mosquito surveillance programme [16] at the University of Notre Dame Environmental Research Center (UNDE) site in the northern Great Lakes region, for the 2024 season and the focal vector species *Coquillettidia perturbans*. This application is a real-data illustration and coherence check: it asks whether the identifiability pattern predicted by Proposition 3 and quantified under a known truth in Section 6 is recognisable in a single real dual-channel dataset. It is not, and is not intended as, a further proof of the fusion payoff—that quantitative claim rests on the controlled simulation, where the truth is known and coverage can be measured. The value of the NEON fit is qualitative: it shows the mechanism survives real, noisy, partially missing data at one site and season.

7.1. Data and dual-channel construction

NEON runs two consecutive ~ 12 -hour trap deployments at each biweekly visit: an *overnight* CO₂ trap (set in the evening, collected the next morning), when mosquito flight activity is high, and a *daytime* trap (set in the morning, collected in the evening), when activity is low. We use the overnight trap as the count channel and the daytime trap as the binary detection channel. Because the two traps run on non-overlapping cycles, they are conditionally independent given the underlying abundance, exactly the assumption of Section 4; yet they are not repeated measurements of one protocol, so the second stream is truly complementary. Let $i = 1, \dots, m$ index the $m = 10$ trapping plots and $t = 1, \dots, T$ the $T = 16$ biweekly time steps (11 April–30 October 2024). Both channels are constructed for the *same focal species*, so that they observe one common latent abundance field: the overnight count is

$Y_{i,t}^{(C)}$ (species-specific *C. perturbans* counts rescaled by the taxonomic subsampling fraction), observed at 149 of 160 plot-times, of which 76 are zeros and the maximum is 4800; the daytime detection $Y_{i,t}^{(P)} \in \{0, 1\}$ records whether *C. perturbans* was found in the day trap, observed at 149 plot-times with 64 detections (detection rate 0.43). The detection channel is defined *species-specifically* on purpose; a coarser “any target taxon present” flag would yield a near-saturated detection rate (≈ 0.77) that does not observe the same latent field as the species-specific count channel and is therefore not used. Trap-hours are near-constant across valid deployments (13.0–14.8 h), so we absorb effort into the detection parameters ρ and κ rather than fitting a separate exposure offset; the general observation model of Section 4 retains explicit effort terms for designs where effort varies materially. Plot land-cover (wetland and forest indicators from NLCD) and standardized elevation form the covariates X_i for the process, and a 2.5 km adjacency graph defines the dispersal neighbourhoods $\mathcal{N}_j$ (mean degree 3.6).

7.2. Model and estimation

The latent field follows the count process of Section 3, with $\log K_i = X_i^\top \beta_K$ linking carrying capacity to land cover, a global persistence ϕ , arrival rate ψ , and a single dispersal fraction δ spread over each source’s neighbours through $g(\cdot; \nu)$. The observation layer is Eqs. (9)–(10) with effort exposures; we anchor the abundance scale to the overnight count by fixing $\rho = 1$, so $N_{i,t}$ is expressed in expected-overnight-catch units and the daytime channel then informs the detection parameters (κ, η) . The latent field is augmented on the log scale ($M_{i,t} = e^{z_{i,t}}$, a log-normal transition with the model’s exact conditional mean and an estimated process dispersion σ_p), which absorbs the strong non-stationarity of the series—near-zero in spring, an order-of-magnitude mid-summer boom, an autumn crash—better than an additive Gaussian; the negative-binomial and Bernoulli observation densities are exact. This log-normal latent field replaces the integer state with a continuous latent intensity: unlike the exact truncated forward filter of Section 6.1, it does not represent the discrete count state and, in particular, cannot place mass exactly at $N = 0$. Accordingly, the NEON analysis should be read as an approximate latent-intensity analysis rather than exact integer-state filtering. We fit the full coupled model by Hamiltonian Monte Carlo (NUTS) with a non-centred parameterisation of the latent field (released as `mos_numpyro_fit.py`), using the priors of Section 6.1 including the informative $\eta \sim \text{Beta}(1, 49)$ false-positive prior. Convergence is assessed by the

split- $\hat{R}$ statistic and effective sample size.

Two properties of this fit should be stated plainly, as they bound what the application can claim. First, the transition surrogate is most strained at the low, near-zero counts that dominate the spring plot-years (of the 149 observed overnight counts, 76 are zeros); its one-step total-variation error (`approx_error_study.py`) is about 0.13 at a conditional mean count of $\mu \approx 2.6$, falling to ≤ 0.04 for $\mu \gtrsim 40$, and depends only on local mean count, not grid size. The exact observation densities limit its effect—at a zero-count plot-year the evidence enters through the exact NegBin and Bern terms—but we flag the low-count regime as the main numerical caveat. Second, and more consequentially, fixing $\rho = 1$ anchors the abundance scale to the count channel by design; the *absolute* scale is only weakly identified from these data (Section 8), so we report the observation parameters and the latent field *in overnight-catch units* and do not claim a data-driven absolute abundance.

Role of fixing $\rho = 1$. In the NEON application, ρ is fixed to 1 to express the latent abundance scale in expected overnight-catch units. This anchoring avoids a weakly identified global abundance–detection ridge, but it also means that the application does not estimate the count-detection fraction ρ and therefore does not exercise Proposition 3(i) on the real data. The simulation in Section 6 is the evidence for joint recovery of the observation vector; the NEON example demonstrates that the species-specific binary stream behaves consistently with the predicted identifiability pattern. An alternative that would exercise Proposition 3(i) on real data is to free ρ under an informative $N(\mu_\rho^0, \sigma_0^2)$ prior, reporting an absolute scale with honest posterior uncertainty; our released fitting code exposes this option (Section 8).

7.3. Results

Table 2 reports posterior medians and 95% credible intervals from a four-chain NUTS fit (all split- $\hat{R} \leq 1.03$). The observation layer retains moderate overdispersion ($r = 4.40$ [1.59, 11.47]), while the bulk of the boom-and-bust volatility is carried by the log-normal process dispersion ($\sigma_p = 2.65$)—consistent with the process-level underdispersion identity (8), which forces any excess variability onto the observation and process-noise layers rather than the latent transition. Because the species-specific daytime channel is informative (detection rate 0.43, not the near-saturated 0.77 of an any-taxon flag), the binary stream now carries real information about (κ, η) : the false-positive floor $\eta = 0.006$ [0.000, 0.031] is estimated near zero, consistent with

	Parameter	Median [95% CrI]	$\hat{R}$
Observation	NB dispersion r	4.40 [1.59, 11.47]	1.03
	Detection intensity κ	0.125 [0.065, 0.222]	1.00
	False-positive floor η	0.006 [0.000, 0.031]	1.01
Process	Persistence ϕ	0.20 [0.03, 0.81]	1.01
	Arrival ψ	0.20 [0.08, 0.46]	1.00
	Dispersal fraction δ	0.52 [0.25, 0.87]	1.01
	Saturation ν	73 [14, 285]	1.01
	Process SD σ_p	2.65 [2.26, 3.11]	1.01
$\log K_i$	Intercept	2.98 [-3.06, 11.12]	1.02
	Elevation (z)	-0.48 [-4.37, 3.44]	1.01
	Wetland	-0.33 [-4.25, 3.80]	1.02
	Forest	-0.68 [-4.63, 3.65]	1.00

Table 2: Posterior summaries for the NEON UNDE dual-channel mosquito fit on the *corrected species-specific* channels ($m = 10$ plots, $T = 16$ biweekly steps, 149 count and 149 detection plot-times). Four-chain NUTS (`mos_numpyro_fit.py`); all split- $\hat{R} \leq 1.03$. The count channel informs the overdispersion r ; the informative binary channel (detection rate 0.43) yields estimates of the false-positive floor η and detection intensity κ consistent with Proposition 3(iii)–(iv), while the absolute abundance scale is anchored by $\rho = 1$ and only weakly identified (wide ν and $\log K$ intercept).

621 a physically clean trap, and the detection intensity $\kappa = 0.125$ [0.065, 0.222]
 622 ($\hat{R} = 1.00$) is reasonably identified through the temporal abundance varia-
 623 tion and the count-channel anchor, consistent with the dual-channel mecha-
 624 nism of Proposition 3(iii)–(iv). This is the main correction in this revision:
 625 with the coherent species-specific channel the fusion behaves as the theory
 626 predicts rather than degenerating to a saturated, uninformative detection
 627 stream. The one identifiability limit that remains is the *absolute abundance*
 628 *scale*: with ρ fixed at 1 and many zero/low count cells, the level of the latent
 629 field is set by the count anchor and prior rather than sharply by the data:
 630 the posterior for the peak total abundance is wide, 32,600 [23,500, 56,500]
 631 (relative width ≈ 1.0 ; Section 8). Land cover shapes carrying capacity in
 632 the expected direction, though the covariate coefficients are not individually
 633 resolved at $m = 10$ plots.

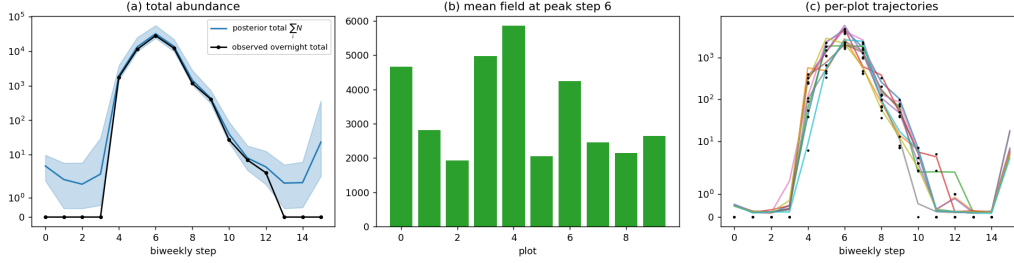


Figure 3: Latent abundance field for *Coquilleltidia perturbans* at NEON UNDE, 2024. (a) Posterior mean total abundance $\sum_i N_{i,t}$ (band: 95%) against the observed overnight-count total, recovering a single mid-summer boom-and-bust; (b) posterior mean latent field $\hat{N}_{i,t}$ across the ten plots at the seasonal peak, higher in the wetland-associated plots; (c) posterior mean per-cell trajectories (lines) with the overnight counts (points) on a symmetric-log scale.

634 The latent field (Figure 3) recovers a coherent seasonal abundance
635 surface—near zero in spring, a sharp mid-summer peak, and autumn
636 decline—with substantial spatial heterogeneity across plots, even though the
637 two raw streams are noisy and partly missing. Its *shape* is well recovered; its
638 absolute level is anchored by $\rho = 1$ rather than sharply pinned by the data
639 (the posterior for the peak total abundance is wide). The model reproduces
640 the key features of both channels in posterior predictive checks (Figure 4):
641 the number of zero counts, the heavy right tail of the overnight counts, and
642 the daytime detection rate all fall within their predictive distributions. As
643 a coherence check, then, the fit behaves as the theory of Section 5 would
644 lead one to expect: the count channel carries the relative abundance and
645 overdispersion, and because the informative species-specific binary channel
646 is not near-saturated (detection rate 0.43), its estimates of the false-positive
647 floor and detection intensity are consistent with the predicted dual-channel
648 identifiability pattern (Proposition 3(iii)–(iv)) rather than degenerating to
649 an uninformative stream. We describe this as consistency with the predicted
650 pattern, not as a proof of recovery: with a single real dataset and no
651 known truth, the fit can corroborate the identifiability structure but cannot
652 measure coverage or error.

653 *What the application does and does not establish..* The strengths of this il-
654 lustration are that the binary channel is species-specific for *C. perturbans*;
655 that its daytime detection rate is 0.43 rather than the near-saturated 0.77 of

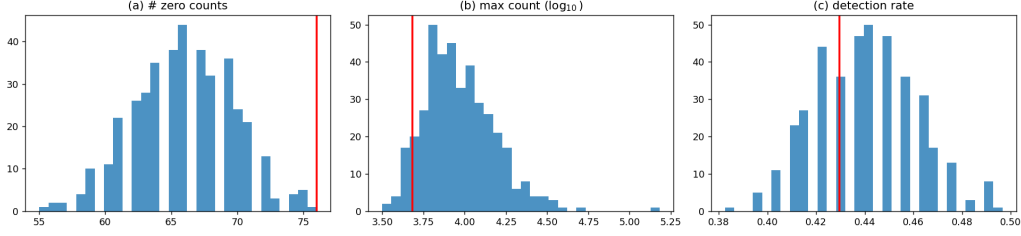


Figure 4: Posterior predictive checks for the NEON UNDE fit. Replicated distributions (histograms) with the observed statistics (red): (a) the number of zero overnight counts, (b) the maximum overnight count ($\log_{10}$ scale), and (c) the daytime detection rate. The observed statistics fall within their posterior predictive distributions, indicating the dual-channel model reproduces the zero-inflation, the heavy right tail of the counts, and the detection rate.

656 a coarse “any-taxon” flag (which does not observe the same latent field as
 657 the count channel, and is therefore not used); that the false-positive floor η
 658 is estimated near zero; that κ is reasonably identified; and that the posterior
 659 predictive checks reproduce the number of zero counts, the heavy right tail,
 660 and the detection rate. Its limitations bound what may be concluded, and
 661 are deliberately foregrounded: the analysis is a *single* NEON site (UNDE),
 662 a *single* season (2024), and only $m = 10$ plots; ρ is fixed at 1, so the count-
 663 detection fraction is not estimated and the recovery of ρ in Proposition 3(i) is
 664 not exercised, and the *absolute* abundance scale is anchored by design rather
 665 than recovered from the data; the series contains many zero and low-count
 666 observations, the regime in which the transition surrogate is most strained;
 667 and the log-normal latent transition is an approximate latent-intensity anal-
 668 ysis rather than exact integer-state filtering (Section 7.2). The application
 669 therefore demonstrates the fusion mechanism and the predicted identifiability
 670 pattern on real data; it does not, and is not intended to, prove full recovery
 671 of every parameter or establish the quantitative payoff, which rests on the
 672 controlled simulation of Section 6.

673 8. Discussion

674 We have treated multi-source count-plus-detection monitoring as a sub-
 675 ject in its own right, with a single message: fusing a count survey and a
 676 binary detection survey improves the practical identifiability of the observa-

677 tion parameters when both streams share a stable count-valued latent abun-
 678 dance process. Concretely, that shared process has a geometrically ergodic
 679 single-cell backbone (Theorem 1) and, under a sub-conservative dispersal
 680 budget $\phi_{\max} + c_{\max} < 1$, a linear conditional-mean drift bound for total abun-
 681 dance on the grid (Proposition 2); it is observed through two conditionally
 682 independent channels whose joint identifiability we characterised analytically
 683 (Proposition 3) and whose practical payoff we quantified in simulation and
 684 then in a real application: a fit to NEON dual-channel mosquito surveillance
 685 data (Section 7) that recovers a coherent seasonal abundance field and obser-
 686 vation parameters whose relative identifiability matches the theory. The divi-
 687 sion of labour between the two empirical sections is deliberate: the value-of-
 688 information claim—that fusion recovers detection parameters neither stream
 689 identifies alone—rests on the controlled simulation of Section 6, where the
 690 truth is known and coverage can be measured, whereas the NEON fit shows
 691 that the predicted identifiability pattern survives real, noisy, partially miss-
 692 ing dual-channel data. With the corrected species-specific detection channel
 693 the informative binary stream yields (κ, η) estimates consistent with the pre-
 694 dicted identifiability pattern. Two points bound how the NEON fit should be
 695 read. First, the evidence for joint recovery of the full observation vector is the
 696 controlled simulation of Section 6, where the truth is known and coverage can
 697 be measured; the NEON fit is a coherence check, corroborating the pattern
 698 on real data rather than proving recovery. Second, because we fix $\rho = 1$ to
 699 anchor the scale to the count channel, the application does not estimate the
 700 count-detection fraction ρ and therefore does not exercise Proposition 3(i) on
 701 the real data. Consequently the one quantity the real data do not pin sharply
 702 is the *absolute* abundance scale: with ρ fixed and many zero and low-count
 703 cells, the level of the latent field is set by the anchor and priors rather than
 704 by the data (the peak-abundance posterior is wide). We are careful not to
 705 over-read the application on this point: it demonstrates the fusion mecha-
 706 nism and estimates consistent with the predicted identifiability pattern, but
 707 not a data-driven absolute abundance.

708 *Implications for monitoring design.* The results speak directly to survey de-
 709 sign. First, a second stream is worth collecting precisely when the count
 710 channel leaves detection and abundance confounded and no strong prior or
 711 replication pins the scale: the value-of-information experiment shows count-
 712 only data cannot estimate the binary detection rate at all, and binary-only
 713 data forecast abundance poorly, while fusion recovers both. Second, temporal

replication is essential for the binary channel: a single presence/absence occasion cannot separate true detection from false positives (Proposition 3(ii), Remark 3), so occupancy-style repeat visits or a multi-year design are prerequisites, not refinements. Third, habitat remote-sensing products should not be entered as detections without a false-positive model; in the mosquito application, land-cover covariates enter the latent process, whereas trap effort enters the observation layer. Fourth, the false-positive rate is weakly identified from short binary series. An informative prior is not merely convenient; if substantially mis-centred, it can bias κ and damage coverage—in the robustness study of Appendix A.1, a mis-centred false-positive prior collapses κ coverage to zero. Thus false-positive priors should be justified from validation data, instrument properties, or external calibration rather than chosen for convenience.

Limitations.. The identifiability results are conditional on positive latent counts and are local (Fisher-information) statements; converting them to global, fully hierarchical identifiability requires process-dependent conditions we do not establish here (Remark 4). Near the unit-root regime the long-distance arrival rate is weakly separated from the count detection fraction, which is why the mosquito fit anchors the abundance scale through the count channel ($\rho=1$) rather than estimating the count-detection fraction and the arrival rate jointly—so the *absolute* abundance scale in the application is set by the anchor rather than the data. An alternative to fixing $\rho = 1$ is to free ρ under an informative $N(\mu_\rho^0, \sigma_0^2)$ prior, which would report an absolute scale *with* honest posterior uncertainty and exercise Proposition 3(i) on the real data; our released fitting code (`mos_numpyro_fit.py`) exposes this option. The false-positive rate η is only weakly identified from a short binary series and benefits from an informative prior; our sensitivity analysis suggests estimates are robust to small η , but datasets with substantial false positives would need a richer proxy model. A further lesson from the application is that the two channels must observe the *same* latent quantity: a coarse “any-taxon” detection flag is near-saturated and does not identify the focal species’ detection parameters, whereas the species-specific channel used here does. Finally, the real application is a single site and season with $m = 10$ plots; extending the same dual-channel likelihood to larger, multi-programme integrations—for example BBS route counts fused with eBird detections over continental grids—is a natural next step.

μ_ρ prior	post. mean μ_ρ	$\bar{\rho}$	$\hat{N}$
$\mathcal{N}(0, 0.5^2)$ (tight)	−0.30	0.426	10.67
$\mathcal{N}(0, 1)$ (main)	−0.39	0.406	10.72
$\mathcal{N}(0.5, 1)$	−0.33	0.420	10.69
$\mathcal{N}(−1.5, 1)$	−0.55	0.369	10.79

Table A.3: Count-only joint estimation of (μ_ρ, K) under varying μ_ρ priors (truth $\mu_\rho = -0.5$, mean $N \approx 10.6$; carrying-capacity prior anchored; exact truncated forward filter). The implied abundance scale $\hat{N}$ is nearly invariant to the prior, so the scale is data-driven (`joint_estimation_study.py`).

750 *Scope of the present article.*.. The manuscript deliberately avoids claims
751 about flexible kernel learning or computational acceleration. Those topics
752 can be studied with many different implementations, but they are secondary
753 to the central statistical issue addressed here: whether the fused observation
754 likelihood separates abundance, count detection, binary detection intensity,
755 and false positives. This focus makes the paper independent and suitable as
756 a statistical-ecology contribution to integrated monitoring design.

757 Appendix A. Supporting simulation studies

758 Two further self-contained studies, released with the paper, support the
759 claims of Sections 6 and 7. Both isolate a single question and neither depends
760 on the choice of dispersal kernel or fitting algorithm, consistent with the
761 paper’s scope.

762 *The count-only abundance scale is data-driven, not prior-driven.*.. Section 6
763 reports that count-only data (regime R1) recover the detection intercept μ_ρ .
764 Table A.3 confirms that this recovery is supplied by the density-dependent
765 dynamics rather than by the prior. Jointly estimating (μ_ρ, K) —and hence
766 the absolute abundance scale—from a single count-only series by an exact
767 truncated forward filter, the posterior mean of μ_ρ stays near the truth (−0.5)
768 and the implied mean abundance $\hat{N}$ is essentially invariant (≈ 10.7 , true
769 value ≈ 10.6) as the μ_ρ prior is moved from $\mathcal{N}(0, 0.5^2)$ to $\mathcal{N}(−1.5, 1)$: a five-
770 fold change in prior location shifts $\hat{N}$ by under 1%. The density-regulated
771 approach to carrying capacity gives the count series a shape that pins the
772 level, so μ_ρ is identified by the data.

Regime (mean N)	Transition	latent-field RMSE	90% cov.
Low (≈ 1.2)	exact	0.879	0.970
	Gaussian	0.877	0.971
	log-normal	0.912	0.637
Application (≈ 9.5)	exact	2.610	0.930
	Gaussian	2.601	0.934
	log-normal	2.606	0.900

Table A.4: Latent-field recovery of particle filters differing only in the one-step transition law, against the true field. The Gaussian surrogate matches the exact filter in both regimes; the log-normal surrogate undercovers in the low-count regime, where it cannot represent $\Pr(N = 0)$ (`low_count_study.py`).

773 *Low-count behaviour of the transition surrogate.* Table A.4 quantifies the
774 caveat of Section 7.2 by comparing bootstrap particle filters that differ only
775 in their one-step transition—the exact Binomial–Poisson law, its moment-
776 matched Gaussian surrogate, and the log-normal surrogate used in the NEON
777 fit—against the true latent field. In the application abundance regime (mean
778 count ≈ 9.5 , the summer plot-years) all three agree closely, with 90% credible-
779 interval coverage 0.90–0.93. In the low-count regime (mean count ≈ 1.2 , the
780 spring and zero-inflated plot-years) the Gaussian surrogate is indistinguish-
781 able from the exact filter, whereas the log-normal surrogate degrades: its
782 coverage falls to 0.64 and, unlike the exact and Gaussian laws, it cannot
783 place mass at $N = 0$ (at $\mu = 1.5$ the exact transition has $\Pr(N = 0) = 0.12$
784 against 0.04 for the log-normal). This locates the numerical error flagged
785 in Section 7.2 precisely at the near-zero counts. The comparison is made in
786 stationary regimes; for the strongly non-stationary mosquito series we retain
787 the log-normal surrogate for the boom-bust reasons given in Section 7.2, and
788 note that an exact truncated-filter or particle treatment is the assumption-
789 free route to eliminating the residual low-count error.

790 *Appendix A.1. Robustness of the value-of-information analysis*

791 Section 6 demonstrates practical recovery of the observation vector under
792 a known truth, but its coverage figures are produced with the demographic
793 process held at its data-generating values and at a moderate abundance,
794 and are therefore optimistic about interval precision. This appendix quan-

795 tifies how far those figures survive the conditions of the real application.
 796 Three further checks (released as `paper2_sim_studies.py`; $R = 20$ repli-
 797 cates, $m = 35$, $T = 18$; exact per-cell forward-filter marginal likelihood in
 798 the same conditionally independent-cell regime as Section 6.1, Laplace 95%
 799 intervals) vary the process nuisance parameters and the abundance regime.
 800 Table A.5 reports the results: estimating the process nuisance parameters
 801 (ϕ, ψ, K) jointly widens the intervals substantially while preserving coverage,
 802 and moving to a low-abundance/high-zero regime likewise widens them, so
 803 the fixed-process, moderate-abundance figures of Section 6 are optimistic
 804 about precision but not about calibration.

Check	Setting	μ_ρ cov. (width)	κ cov. (width)
Process nuisance	(ϕ, ψ, K) fixed at truth	1.00 (0.22)	0.95 (0.020)
	(ϕ, ψ, K) estimated jointly	0.95 (1.49)	0.95 (0.129)
Abundance regime	moderate ($\bar{N} \approx 7.5$, 24% zeros)	0.90 (0.23)	1.00 (0.020)
	low ($\bar{N} \approx 1.5$, 72% zeros)	0.95 (0.43)	0.95 (0.080)

Table A.5: Coverage of nominal 95% intervals ($R = 20$; Monte Carlo standard error ≤ 0.11). Coverage is maintained when the process is estimated jointly and in the low-abundance/high-zero regime, but the intervals widen sharply (the κ interval is $\sim 6\times$ wider with the process estimated, and $\sim 4\times$ wider at low abundance): the fixed-process, moderate-abundance figures of Section 6 are therefore optimistic about *precision*, though not about *calibration*.

805 A third check varies the false-positive prior (true $\eta^* = 0.15$). With a
 806 correctly informative prior (Beta(3, 17)) both η and κ are recovered with
 807 near-nominal coverage; under a *diffuse* prior (Beta(1, 1)) the η coverage falls
 808 to 0.80 and the joint (κ, η) estimate becomes unstable; under a *mis-centred*
 809 prior (Beta(1, 49), mean 0.02) the resulting η bias propagates to κ , whose cov-
 810 erage collapses to 0.00. The false-positive rate is thus only weakly identified
 811 from a short binary series and is materially prior-dependent—an informative,
 812 correctly-centred prior is a modelling requirement, not a convenience, and a
 813 mis-specified one is actively harmful. This sharpens the limitation already
 814 noted in Section 8.

815 Data and Code Availability

816 The NEON mosquito surveillance data (overnight CO₂-trap counts and
817 daytime-trap detections) analysed in Section 7 are openly available from
818 the National Ecological Observatory Network data portal [16]. The North
819 American Breeding Bird Survey and eBird Basic Dataset referenced as a
820 natural extension are available, respectively, from the USGS BBS data re-
821 lease [17] and from eBird after a data request [18]. The full analysis code is
822 released as a versioned repository (<https://github.com/xulinpan/scabm>)
823 archived at the public DOI [10.5281/zenodo.21211674](https://doi.org/10.5281/zenodo.21211674), snapshotted at the git
824 tag used for this article and accompanied by an environment specification for
825 exact reproduction. The repository contains the reference model implementa-
826 tion (`sc_abm_model.py`); the value-of-information simulation and estimator
827 of Section 6.1; the process-validation experiments of Section 3.3; and the
828 supporting studies of Appendix Appendix A (`joint_estimation_study.py`,
829 `low_count_study.py`) together with the transition-approximation charac-
830 terisation of Section 7.2 (`approx_error_study.py`). All scripts use fixed
831 pseudo-random seeds so that every figure and table is reproducible from a
832 clean checkout.

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